SIMATS ENGINEERING

ASSESSMENT TOOL 4

COURSE NAME: Artificial Intelligence COURSE CODE: CSA1711

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 3

Total Marks:30

Annexure A

Dry Run Evaluation

Code of Conduct:

I D.Balaji (192512368) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D.Balaji

1. Breadth-First Search (BFS)

Given Graph

```
      A
     /  \
    B     C
   / \   / \
  D   E F   G
```

BFS Process

Iteration	Queue (Before Removing)	Visited Node
1	A	A
2	B, C	B
3	C, D, E	C
4	D, E, F, G	D
5	E, F, G	E
6	F, G	F
7	G	G

Queue After Each Iteration

Iteration	Queue After Expansion
1	B, C
2	C, D, E
3	D, E, F, G
4	E, F, G
5	F, G
6	G
7	Empty

Answers

1. BFS Traversal Order

A → B → C → D → E → F → G

2. Queue Contents After Each Iteration

Iteration 1 : B, C

Iteration 2 : C, D, E

Iteration 3 : D, E, F, G

Iteration 4 : E, F, G

Iteration 5 : F, G

Iteration 6 : G

Iteration 7 : Empty

2. Depth-First Search (DFS)

DFS Traversal (Left Child First)

```
      A
    /   \
   B     C
  / \   / \
 D   E F   G
```

DFS Process

Iteration	Stack	Visited Node
1	A	A
2	B, C	B
3	D, E, C	D
4	E, C	E
5	C	C
6	F, G	F
7	G	G

DFS Traversal Order

A → B → D → E → C → F → G

3. 8-Puzzle Problem Using Breadth-First Search (BFS)

Initial State

1 3 6

5 0 2

4 7 8

Goal State

1 2 3

4 5 6

7 8 0

Since BFS explores level by level, the search tree grows rapidly. The goal state is **not reached within only five iterations** for this puzzle, so a complete BFS would require many more expansions.

First Five Iterations

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1	1 3 6 / 5 0 2 / 4 7 8	Initial	4 successor states
2	First generated state	4 states	New unexplored states added
3	Second generated state	Remaining states	New states added
4	Third generated state	Remaining states	New states added
5	Fourth generated state	Remaining states	New states added

Possible Successors of Initial State

Blank (0) is in the center, so it can move **Up, Down, Left, Right**.

Move Up

1 0 6

5 3 2

4 7 8

Move Down

1 3 6

5 7 2

4 0 8

Move Left

1 3 6

0 5 2

4 7 8

Move Right

1 3 6

5 2 0

4 7 8

Thus, **4 new states** are generated in the first expansion.

Answers

1. Which node is expanded in each iteration?

- Iteration 1 → Initial State
- Iteration 2 → First generated state
- Iteration 3 → Second generated state
- Iteration 4 → Third generated state
- Iteration 5 → Fourth generated state

2. How many states are generated in total?

- First expansion generates **4 states**.
- The total number of generated states until the goal depends on the complete BFS search and is much larger than 4. It cannot be determined without expanding the entire search tree.

3. Complete Solution Path

The solution path cannot be determined manually in only five iterations because the given initial state is several moves away from the goal. A complete BFS search must continue until the goal state is found.

4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

Answer:

- BFS explores states **level by level**.
- Every move has the **same cost (1 move)**.
- It visits all states requiring **n moves** before exploring states requiring **n + 1 moves**.
- Therefore, the **first time the goal state is reached, it is guaranteed to be the shortest possible path**.
- Hence, BFS always finds the **optimal (minimum-move) solution** in an unweighted 8-puzzle problem.